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Purpose/Objective(s): There is growing evidence that stem cells exist in tumors and contribute to treatment resistance. In this study, we have utilized low passage head and neck tumor (H&N) cell lines, isolated cancer stem cells (CSC) using different methods and subjected them to gene expression analysis. The goal is to find new biomarkers for stem cells for prognostic use and to discover new targets for treatment.

Materials/Methods: A panel of five (UT14 UT16A, UT24A, UT30 and UT33) low passage H&N cancer lines was studied. CSC (and non CSC) were isolated from these by high-speed cell sorting using two different flow cytometry techniques 1) Hoechst 33342 “side” population and 2) the Aldefluor assay. RNA was extracted from the sorted populations for gene expression analysis and further experimental studies of radiosensitivity. Differentially expressed genes were determined by utilizing Affymetrix Human Exon 1.0 ST microarrays and subsequent analysis in Partek’s Genomics Suite.

Results: The two methods of isolation identified different proportions of “stem” cells. The “side” population was less prevalent and ranged from 0.3 to 3.7% in the 5 cell lines whereas Aldefluor positive cells ranged from 6 to 31.4%. Only UT33 had a similar proportion of cells (3.7 vs. 6%) by the two methods. In the analysis so far using ANOVA with a significance of $p \leq 0.05$ and a fold-change cutoff of 1.5, 83 differentially expressed genes were identified by comparing “side” population to non-“side” population samples. of the 45 genes that were up-regulated, 22 were associated with histones.

Conclusions: The high incidence of histone gene overexpression in “side” population cells is consistent with the known involvement of chromatin packaging in the maintenance of the stem cell phenotype and the involvement of histones in DNA repair may relate to the increased radioresistance of stem cells. Further studies on the epigenetic regulation of histones in stem cells may provide future avenues for therapeutic targets to be used in conjunction with chemoradiation.

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282 Fast and Accurate Verification of Proton Range, Position, and Intensity in IMPT with a 3D Detector System

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Purpose/Objective(s): Intensity-modulated proton therapy (IMPT) using scanned proton spots relies on the delivery of a large numbers of spots to shape the dose distribution in a highly conformal manner. We have developed a 3D system based on liquid scintillator (LS) to evaluate the spot position, intensity (dose) and range (energy) of proton spots in real time.

Material/Methods: The LS is a tissue equivalent material emitting scintillation light when irradiated by ionizing radiation. The LS detector system is comprised of a 20×20×20 cc volume of LS in a light tight enclosure to prevent against ambient light contamination. The LS detector is rigidly coupled to a CCD camera. This camera continuously acquires images of the light emitted by the LS therefore allowing a quantitative characterization of the proton beam. Image acquisitions are triggered by a signal from the proton accelerator. Irradiations were conducted on a proton spot scanning beam line for which one monitor unit (MU) is defined as 1 c Gy in the middle of a 10 cm spread-out Bragg peak for a field of 10 cm by 10 cm with a range of 30.6 cm in water. Scintillation emission is expected to be dependent on the proton energy due to the quenching effect. Therefore, irradiations were done to evaluate the magnitude of this effect and test correction strategies.

Results: **Range:** Irradiations with protons of known ranges were used for calibration. We found that one centimeter of LS was equivalent to 0.87 cm of water. Measurement of proton range with the LS detector system was found to be within 0.1 mm (0.3 mm maximum error) for protons with nominal range between 6.6 cm and 10.5 cm. Moreover, proton range measurements did not require a quenching correction. **Position:** Measurement of lateral position of proton spots was found to be accurate to within 0.4 mm (0.6 mm maximum error). Imaging artifacts such as vignetting did not impact our ability to measure spot position. However, it was necessary to correct for image distortion caused by the objective lens. **Intensity:** The system was sensitive enough for accurate measurements of spot intensity delivered at the lowest (0.005 MU) and highest (0.04 MU) dose per proton spot. The precision of the detector was 3.3% and 0.5% for 0.005 MU and 0.04 MU, respectively. Quenching had a strong effect on relative spot intensity measurement, however, a Monte Carlo based correction could be used to completely account for this effect.

Conclusions: Our LS detector system is well suited for fast and accurate verification of proton treatments using scanned proton beams such as IMPT. Proton spots range and position could be measured with sub-millimeter accuracy. The system was sufficiently sensitive to measure individual proton spots in real-time. This new device has potential to both improve and simplify the quality assurance process of IMPT treatments.

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283 OPTIS2: A Second Generation Horizontal Beam Line for Ocular Proton Therapy using a 250 MeV Cyclotron - First Patients Treated 25 Years after Start of the OPTIS Program at Paul Scherrer Institute

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Purpose/Objective(s): In 1984 proton therapy was introduced to Europe with start of the OPTIS program at Paul Scherrer Institute. A prototype “low energy” 72 MeV cyclotron provided protons of sufficient energy to treat ocular lesions. To date more than 5000 uveal melanoma patients have been treated with local tumor control of 97% at 5 years. To continue the program using the horizontal beam line connected to “high energy” 250 MeV cyclotron COMET a new facility (OPTIS2) was designed and constructed. As the current generation of high energy cyclotrons provide a beam current at 70 MeV that is dramatically lower than